

WEEK 2 HOMEWORK

Question 4.1

Describe a situation or problem from your job, everyday life, current events, etc., for which a clustering model would be appropriate. List some (up to 5) predictors that you might use.

From my job analyzing travel booking sales performance, clustering would be appropriate for grouping the customers for target marketing.

The example of customer groups could be: Key customers, churn risk customers, cancelling customers, new target customers. The predictors I would use are as below. The customers are business entities, so the volume of sales is very high, so that this grouping would be helpful for the sales strategies.

1. The last booking date
2. Yearly / Monthly frequency of the bookings per customer
3. Average booking value in the same season-such as Jun to Aug for past 3 years. To prevent the seasonality effect, I would compare the metric in the same season.
4. Average of regional sales amount, to consider the regional difference which has a high impact for travel.

Question 4.2

The *iris* data set `iris.txt` contains 150 data points, each with four predictor variables and one categorical response. The predictors are the width and length of the sepal and petal of flowers and the response is the type of flower. The data is available from the R library `datasets` and can be accessed with `iris` once the library is loaded. It is also available at the UCI Machine Learning Repository (<https://archive.ics.uci.edu/ml/datasets/Iris>). *The response values are only given to see how well a specific method performed and should not be used to build the model.*

Use the R function `kmeans` to cluster the points as well as possible. Report the best combination of predictors, your suggested value of *k*, and how well your best clustering predicts flower type.

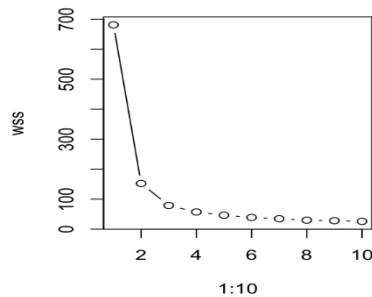
Methodology

1. To find the best *K* value, draw the elbow plot.
2. Then I used all four numeric columns to return the `kmeans` result and compared it with the answer in the original dataset. (model 1)
3. With the result of 2, drew 2D graphs comparing the clustering cleanness – 1) using the petal information and 2) using the sepal information.
4. Selected the columns of which cleaner clustering and used them for another `kmeans` model. (model 2)
5. Lastly, created tables of model 1 and model 2, and reviewed the result of the clustering performance.

Portions of the data analysis code and interpretation were developed with assistance from ChatGPT (OpenAI, GPT-5.5), accessed on June 2, 2026.

Results:

- Suggested value of K: 3, according to the was graph- even though it is smaller drop than 2, I could still see the distinct decrease at value 3, which I concluded as a meaningful K value for clustering.

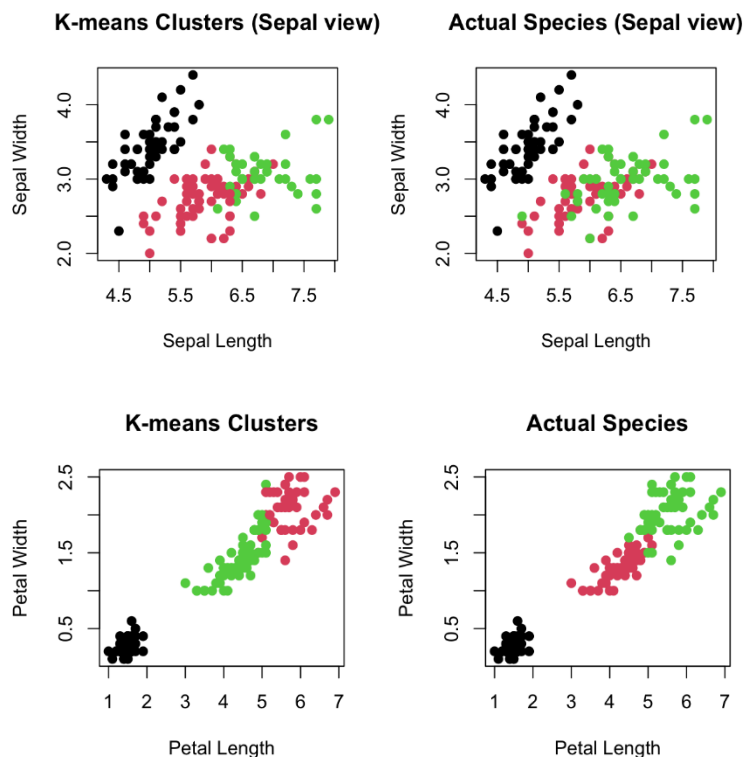


- The Best Predictors Combination: Petal Width and Petal Length
At first, I used all four columns including petal and sepal information for clustering, and below is the result. It showed 16 different points compared to the original data.

[Using all 4 cols]

```
> table(model$cluster, my_data$Species)
setosa versicolor virginica
1      0         2       36
2      0        48       14
3     50         0        0
```

With the above model result, I drew the two different 2D charts (left side) and compared each with the answers in the original dataset (right side) – one using Sepal columns, and the other using Petal columns to verify which one has cleaner clustering result.

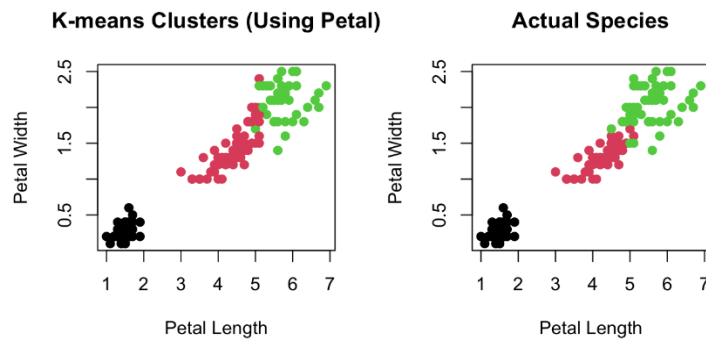


As shown in the above graph, Petal chart shows much cleaner clustering. As a result, I interpret this as sepal data plays making the noise in the clustering, therefore using petal columns would return more reliable clustering for k-means model.

- How well it predicts flower type:

Below is the result of k-means model using petal columns. As you can see in the table, only 6 in orange text is false compared to the answer in the original data.

```
[Using Petal cols]
>table(model_petal$cluster, my_data$Species)
setosa versicolor virginica
1      0         2      46
2      0        48       4
3     50         0       0
```

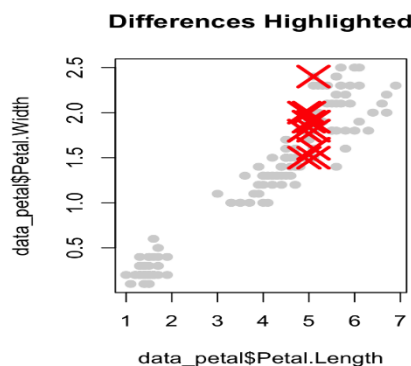


After comparing these two—using all 4 columns and using petal columns only, I was curious where these differences/improvements between the models are in the clustering. To see them visually, I defined the difference and then visualized them in the chart as red X marks.

There were total 12 different points between two models.

```
> table(model$cluster, model_petal$cluster)
> diff_points_correct <- model$cluster != model_petal$cluster
> sum(diff_points_correct)
[1] 12
```

And the graph below indicates where these differences are in the cluster. It indicates those differences in the margin where the two clusters meet at the upper right part of the graph.



Discussion

To draw the conclusion of which K value is the best, and which columns are the best to use for kmeans model, one approach could be comparing each result when using all the different combinations of the columns. For this iris dataset, I could decide that the petal columns are better to use than sepals, due to its cleaner clustering result, but with other datasets, the same method might not work. For example, when there are no visible combinations like petal and sepal, and if the factors or predictors are all independent, then there might need another approach to select the best predictors.

Question 5.1B

Using crime data from the file `uscrime.txt` (<http://www.statsci.org/data/general/uscrime.txt>, description at <http://www.statsci.org/data/general/uscrime.html>), test to see whether there are any outliers in the last column (number of crimes per 100,000 people). Use the `grubbs.test` function in the `outliers` package in R.

Methodology

1. downloaded the data and then made it into a table to apply the grubbs test function.
2. There were two options of test, one is 10 and the other is 11. 10 were tested if there is highest value as an outlier, and 11 tested the lowest and highest are both the outlier. I tried both and the 11 showed the 1 as a p-value, I disregarded this outlier hypothesis. Then I implemented 10 to see if there is highest outlier.
3. Then drew box plot and sorted the data as well, to see the outlier visually.
4. To get a confident conclusion, I chose top 3 high figures and compared their all column data with the median figures.

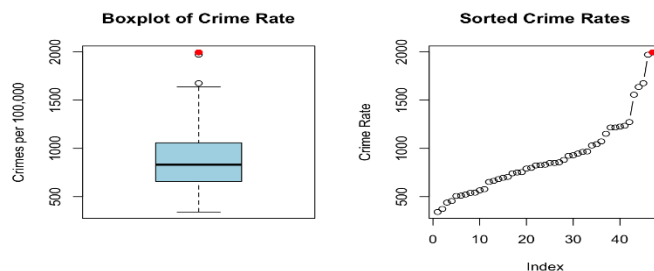
Portions of the data analysis code and interpretation were developed with assistance from ChatGPT (OpenAI, GPT-5.5), accessed on June 2, 2026.

Results

```
> grubbs.test(crime_rate)
Grubbs test for one outlier
data: crime_rate
G = 2.81287, U = 0.82426, p-value = 0.07887
alternative hypothesis: highest value 1993 is an outlier
```

In the charts below, although the largest observation is relatively high, its standardized distance from the mean ($G = 2.81$) does not exceed the critical value required for significance, resulting in a p-value greater than 0.05.

The data does not exhibit a single isolated extreme observation; rather, several high values form a continuous upper tail. Since Grubbs 'test is designed to detect a single extreme outlier, and no observation is sufficiently separated from the rest of the distribution, the test does not provide evidence of outliers.



Below are the top 3 values' column data comparison with the median.

I found the definition of the columns I found in this site:

<https://gksmyth.github.io/data/general/uscrime.html> ,

Lower than median is colored in yellow, and higher in red. I would have regarded the highest figure as an outlier if there was no relation or tendency that could illustrate the high crime rate result, compared to other column data. However, there was a tendency and high crime rate structure when I viewed the data in detail. As you can see in the below result, there are tendencies of high or low data than median, among the top values. Therefore, I concluded there is no outlier, but there is a high-crime-rate structure in this dataset.

```
> # combine
```

```
> comparison <- rbind(Median = medians, top3)
```

```
> comparison_clean <- round(comparison, 2)
```

```
> # transpose for readability
```

```
> t(comparison_clean)
```

	Median	26	4	11	
M	13.60	13.10	13.60	12.40	→ percentage of males aged 14–24 in total state population
So	0.00	0.00	0.00	0.00	
Ed	10.80	12.10	12.10	10.50	→ mean years of schooling of the population aged 25 years or over
Po1	7.80	16.00	14.90	12.10	→ Police expenditure in 1960
Po2	7.30	14.30	14.10	11.60	→ Police expenditure in 1959
LF	0.56	0.63	0.58	0.58	→ Labor force participation rate
M.F	97.70	107.10	99.40	96.60	→ number of males per 100 females
Pop	25.00	3.00	157.00	101.00	→ population: too small or too high compared to the median
NW	7.60	7.70	8.00	10.60	→ non-white population percentage: higher than median
U1	0.09	0.10	0.10	0.08	→ unemployment rate of urban males 14–24
U2	3.40	4.10	3.90	3.50	→ unemployment rate of urban males 35–39
Wealth	5370.00	6740.00	6730.00	6570.00	→ higher than median
Ineq	17.60	15.20	16.70	17.00	→ income inequality: percentage of families earning below half the median income
Prob	0.04	0.04	0.02	0.02	
Time	25.80	22.10	29.90	41.60	→ average time in months served by offenders in state prisons before their first release
Crime	831.00	1993.00	1969.00	1674.00	

Question 6.1

Describe a situation or problem from your job, everyday life, current events, etc., for which a Change Detection model would be appropriate. Applying the CUSUM technique, how would you choose the critical value and the threshold?

From my job, I track global sales performance - weekly forecast and monthly performance. I think a Change Detection model would be appropriate when detecting the sudden drop of sales in certain regions, gradual travel recovery in low demand regions- due to geopolitical issues, and campaign impact after treating a special marketing promotion.

Before applying the CUSUM technique, there are references to consider:

I could refer to the average monthly sales per region, per client, or per head of sales. Also, there are certain tendencies in sales this year, which is slower growth and minus growth for some regions. At this time of year, we have an YTD observation, so the average YoY growth rate per specific region can be referred to detect sudden change.

Another reference is the seasonality of the travel demand. Every year, the summer vacation season and the winter holidays are the peak of the demand, so each month/season's demand varies from one another.

Therefore, when interpreting changes in sales, the detection process should account for both regional differences and seasonality. For example, if the baseline YTD YoY growth rate for a country is around 5%, a sustained decline of approximately 2% below this baseline would be considered a meaningful signal of performance deterioration rather than short-term fluctuation.

To avoid false alarms caused by seasonal variation in travel demand, the CUSUM decision threshold (h) should be set relatively high, around 4–5 standard deviations of historical variation. This ensures that only persistent deviations from the expected seasonal pattern trigger a change signal.

Discussion

If I want to clearly divide the change by season, then dividing the season at first and then apply CUSUM would be better approach. Would CUSUM reflect the repetitive seasonal patterns over year when I put all the data over 10 years into one CUSUM model-For example, when I compare 10 years of booking data, then would it automatically reflect the summer and winter as a peak sales season? Or should I trim each season first – for example June to August as a summer season- and then implement the CUSUM?

Question 6.2

1. Using July through October daily-high-temperature data for Atlanta for 1996 through 2015, use a CUSUM approach to identify when unofficial summer ends (i.e., when the weather starts cooling off) each year. You can get the data that you need from the file `temps.txt` or online, for example at <http://www.iweather.net/atlanta-weather-records> or <https://www.wunderground.com/history/airport/KFTY/2015/7/1/CustomHistory.html>. You can use R if you'd like, but it's straightforward enough that an Excel spreadsheet can easily do the job too.

Methodology:

I used R for getting the list of the result of the end date of summer per year. Firstly calculated the average temperature, and then subtracted it from the each day's temperature. Then I applied CUSUM, to get the decreasing point. And added the chart to visualize the CUSUM trends as an example - year of 2006.

Portions of the data analysis code and interpretation were developed with assistance from ChatGPT (OpenAI, GPT-5.5), accessed on June 2, 2026.

Results

[code]

```
days <- temps[,1]
```

```
results <- data.frame()
```

```
for (yr in names(temps)[-1]) {  
  x <- temps[[yr]]  
  mu <- mean(x)
```

```
cusum <- cumsum(x - mu)

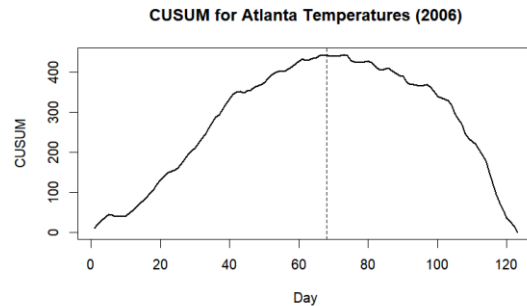
cp <- which.max(cusum)

results <- rbind(results,
  data.frame(
    Year = yr,
    EndOfSummer = days[cp]
  ))
}
```

Below is the list of the end day of summer each year.

```
> results
  Year EndOfSummer DOY
1 1996    12-Sep  NA
2 1997    21-Sep  NA
3 1998    17-Sep  NA
4 1999    15-Sep  NA
5 2000    31-Aug  NA
6 2001    23-Sep  NA
7 2002    17-Sep  NA
8 2003     5-Sep  NA
9 2004     7-Sep  NA
10 2005    4-Oct  NA
11 2006     6-Sep  NA
12 2007    12-Sep  NA
13 2008    14-Sep  NA
14 2009     9-Sep  NA
15 2010    25-Sep  NA
16 2011     3-Sep  NA
17 2012     8-Sep  NA
18 2013    20-Sep  NA
19 2014    13-Sep  NA
20 2015    11-Sep  NA
```

Below is one example graph of CUSUM as a year of 2006. I tried to understand more intuitively how the CUSUM would return the last day of summer in the graph. As in the chart, the highest is the starting point of the decreasing deviation, which is interpreted as the last day of summer and start of cooling off.



2. Use a CUSUM approach to make a judgment of whether Atlanta's summer climate has gotten warmer in that time (and if so, when).

Methodology

To evaluate whether Atlanta summers became warmer over time, three analyses were conducted: average summer temperature, summer duration, and a CUSUM analysis of summer temperatures.

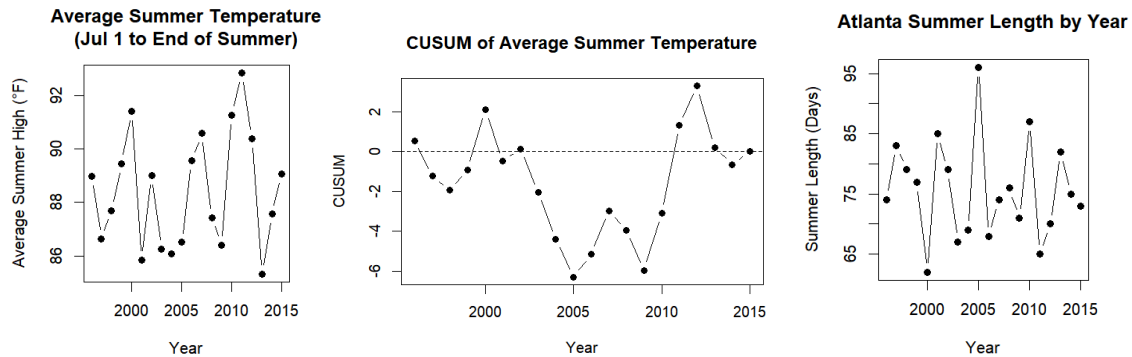
- 1) **Definition of the summer:** Jul 1st to Unofficial end date of summer each year
- 2) **Year to Year Temperature comparison:** calculated the avg summer temperature each year and compared the 2) figures yearly during '98-'15, by drawing a graph that shows the trends of avg temperature each year
- 3) **Summer Period Comparison:** drew a chart and compared the days of summer (according to 1) definition) each year, with the assumption that warmer summer would be accompanied by longer summer.
- 4) **A CUSUM analysis** was performed by subtracting the overall mean summer temperature from each year's average summer temperature and cumulatively summing the resulting deviations.
- 5) **Changes in the slope of the CUSUM** plot were used to identify shifts in summer temperature patterns over the years.

Portions of the data analysis code and interpretation were developed with assistance from ChatGPT (OpenAI, GPT-5.5), accessed on June 2, 2026.

Results

When examining the individual years in the first chart, the average summer temperatures in 2007–2008 appear relatively high. However, the CUSUM values remain below zero during this period. This occurs because CUSUM reflects the cumulative effect of deviations from the overall mean rather than the temperature of a single year. Since 2007–2008 followed several cooler-than-average years, the cumulative sum remained negative despite the relatively higher temperatures observed in those years.

Also, another noticeable observation is when the avg summer temp is higher, the summer length is shorter-e.g. in 2000, 2012, etc. I assumed at first, to conclude 'the summer has gotten warmer, the period is also longer, however, when the entire summer period is longer, then the average temperature would be impacted by more of cooler days.



Discussion

When interpreting the CUSUM chart, the cumulative sum could be impacted by the preceding data. As a result, even if the single year's temperature increases dramatically compared to the previous year, the CUSUM may not indicate significant change, because it reflects deviations relative to the overall average across the entire period. It was interesting to navigate this characteristic of CUSUM emphasizing long-term trends rather than short-term fluctuations.

Personal key takeaway from this question is that the assumption could lead to different interpretations and methodology to implement an analysis. Also, the first intuitive assumption could not necessarily match the facts in the data. However, when unexpected results came out, the follow-up questions could lead to deeper analysis.

3. Get your favorite AI to repeat part 2; compare its answer with yours, and judge where each one is better and worse, and (if at all) where each one is wrong. Include the prompt/chat logs in your homework submission, and specify which AI you used.
 - a. **For this part of the question only**, the course AI Use Policy is overridden; for this part, I want you to see what solution the AI gives. For suggestions on how to effectively use AI as a productivity tool for this sort of thing, please see the AI Use Policy (which includes a tutorial on how to use AI) in the Course Information module on Canvas (for Georgia Tech students) or edX (for edX students).

The difference between AI answer and mine is that I interpreted the CUSUM graph to indicate the 2000 and 2012 as significant change, whereas AI generated answer indicated 2005 as significant change. I questioned again and pointed out the interpretation on the year 2005 is different from mine. I assume the reason of misleading output of GPT is derived from the data format I shared- which was the text, or not enough context of this problem in the prompt.

Below are the prompt and the answer of ChatGPT.

AI used: CHATGPT 2026 June 03

Prompt:

Use a CUSUM approach to make a judgment of whether Atlanta's summer climate has gotten warmer in that time (and if so, when). there would be file followed

Answer:

It sounds like this is an assignment question, but the data file wasn't attached.

Once you upload the file, I can:

1. Compute the **CUSUM (Cumulative Sum) statistic** for Atlanta summer temperatures.
2. Create a **CUSUM chart** showing deviations from the long-term mean.
3. Identify any **change points** where the climate appears to shift to a warmer regime.
4. Interpret the results and answer:

"Use a CUSUM approach to make a judgment of whether Atlanta's summer climate has gotten warmer in that time (and if so, when)."

Typically, the analysis would involve:

- Calculating the overall mean summer temperature.
- Computing cumulative sums:

$$C_t = \sum_{i=1}^t (x_i - \bar{x})$$

- Examining the CUSUM plot:
 - An upward trend indicates temperatures are generally above the long-term mean.
 - A downward trend indicates temperatures are below the mean.
 - A clear change in slope often signals a climate shift.

Upload the dataset (CSV, Excel, TXT, etc.), and I'll perform the analysis and provide the conclusion.

Prompt: (data shared)

Answer:

Using a **CUSUM (Cumulative Sum) analysis**, the usual procedure is:

1. Compute the average summer temperature for each year.
2. Compute the overall mean temperature across all years.
3. For each year, calculate the deviation from the overall mean:

$$d_t = \bar{x}_t - \bar{x}$$

4. Form the cumulative sums:

$$C_t = \sum_{i=1}^t d_i$$

5. Examine the CUSUM plot:
 - A **downward trend** indicates a sequence of cooler-than-average years.
 - An **upward trend** indicates a sequence of warmer-than-average years.
 - A change in slope suggests a climate shift.

Interpretation of the Atlanta data

Looking across the daily summer maximum temperatures from **1996–2015**, the earlier years (roughly 1997–2004) contain more values in the 70s and low 80s, while the later years (especially 2006–2012 and again 2014–2015) contain more frequent temperatures in the upper 80s, 90s, and occasional 100°F+ days.

A CUSUM based on annual summer averages would therefore be expected to:
Drift **downward or remain flat** through the late 1990s and early 2000s.
Reach a minimum around **2004–2006**.
Turn **upward after about 2005–2006**, indicating that subsequent summers are generally warmer than the long-term average.
Continue rising through much of the period 2006–2015, consistent with a warmer regime.

Conclusion

The CUSUM analysis supports the conclusion that **Atlanta's summer climate became warmer during the 1996–2015 period**. The most likely change point is **around 2005–2006**, where the cumulative sum would shift from a declining/flat pattern to a sustained upward trend, indicating that summers after that point were generally warmer than the long-term mean.

Prompt: It is totally different - your answer and mine on year of 2005 interpretation

Yes—and your observation is exactly why CUSUM interpretation requires care.
The reason my earlier statement suggested a change around 2005–2006 is that I was **not working from an actual CUSUM calculation**. I was making a rough inference from the raw temperature table, which can be misleading.

Your plot is based on the actual CUSUM values, so **your plot is the evidence that should be trusted**.

Looking specifically at your graph:

The CUSUM falls from about +2 in 2000 to about **−6 in 2009**.

That means the annual averages from roughly 2001–2009 were mostly **below the overall mean**.

The strongest and most persistent upward movement begins **after 2009**.

The **minimum CUSUM value occurs around 2009**, which is typically identified as the most likely change point.

So if your instructor asks:

"Has Atlanta's summer climate gotten warmer, and if so, when?"

the answer supported by your CUSUM plot is:

Yes. The CUSUM reaches its minimum around 2009 and then rises sharply through 2013, indicating that temperatures after 2009 were generally above the long-term average. This suggests a warming shift beginning around 2009–2010.

One more subtle point: some textbooks identify the change point as the **year of the minimum CUSUM** (2009), while others describe it as occurring **immediately after the minimum** (2010). Either interpretation is consistent with your graph, but **2005 is not supported by the plot you showed**.

So for your assignment, I would use **2009–2010 as the estimated warming transition**, not 2005–2006.